

Influence-Inspired Spectral Rotations for Extreme Low-Bit LLM Quantization

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Abstract

We apply the influence-adaptive Walsh geometry developed in a companion theory paper [1] to the practical problem of extreme low-bit weight-only quantization of large language models. The core idea is one identity transformation: rotate each linear layer’s weight matrix into a Walsh-Hadamard spectral basis and re-scale its columns by per-coordinate spectral activation energy before handing off to an off-the-shelf reconstruction-error quantizer (Intel auto-round). This transformation is exactly invariant at full precision, but biases the per-group integer rounding budget toward channels that carry more spectral signal. Empirically, on four pretrained decoder-only models from two architecture families spanning 135 M to 1.5 B parameters, the recipe reduces wikitext-2 perplexity by 15–58% relative to vanilla auto-round at W2A16 (g=64); a TinyLlama-1.1B auxiliary run is reported where the vanilla baseline was hardware-limited. Three follow-on extensions handle architectures the basic recipe initially failed on: a per-head PCA matrix- Γ replacement of q_norm/k_norm for Qwen3-style attention (PPL 136.76 $\rightarrow$ 88.99 on Qwen3-0.6B), a SO(2)-pair PCA that commutes with RoPE for non-norm architectures (PPL 36.93 $\rightarrow$ 21.84 on Qwen2.5-1.5B at W2), and an MoE-aware input-side absorption fix identified by synthetic architectural fuzzing of Laguna-style fused-expert layouts. A bit-width ablation at W2 vs. W4 shows the redistribution payoff scales with the per-channel quantization-noise budget, falling within the evaluation noise floor at W4 — consistent with the Schur-convexity intuition from the companion theory paper. We do not claim a tight formal connection between the Boolean-function influence of the theory paper and the real-valued WHT activation energy used here; the link is intuitive (per-coordinate Walsh-basis spectral mass) and the empirical PPL improvements are substantial, but the transfer of the majorization argument from the Boolean to the real-valued setting is qualitative.

1 Introduction

A companion paper [1] (henceforth **the BBT theory paper**) develops an influence-adaptive Banach geometry on the Walsh-Hadamard butterfly factorization. The central object is a function-dependent contraction invariant

$$\mu(f) = \prod_{\ell} 2^{-\text{Inf}_{\ell}(f)/(1+\text{Inf}_{\ell}(f))}$$

on Boolean functions $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$, which the theory paper proves is strictly Schur-convex in the influence vector (concentrated influence yields larger μ , smaller margin in the Walsh basis, and harder ternary polynomial-threshold representation).

This paper takes the same Walsh-basis intuition — coordinates with more spectral mass deserve more quantization grid — and applies it to extreme low-bit weight-only quantization of pretrained LLMs. We do *not* attempt to formally extend the Boolean Schur-convexity argument to real-valued layers; the theory paper proves things about Boolean influence and its ternary representations, while this paper makes empirical claims about how an analogously-defined real-valued spectral-energy quantity reweights INT2/INT4 reconstruction error in a way that improves wikitext-2 perplexity. The connection is qualitative: *the same Walsh-basis structure that gives ternary-PTF synthesis a meaningful difficulty diagnostic at the Boolean end gives a meaningful per-channel weighting signal at the real-valued end.*

Contributions.

- (i) A drop-in spectral-basis pre-quantization transformation: WHT-rotate a linear layer’s weight, scale its columns by per-coordinate WHT activation energy, hand off to auto-round. Math-invariant at full precision (verified to relative error $< 5 \times 10^{-7}$); biases per-group INT2/INT4 rounding toward high-energy channels.
- (ii) Empirical validation across four main LLMs (SmolLM-135M/360M, Qwen2.5-0.5B/1.5B) at W2A16: 15–58% wikitext-2 PPL improvement, with the largest wins on models whose vanilla W2 baselines are most degraded; a TinyLlama-1.1B auxiliary data point is reported separately.
- (iii) Three architectural extensions that transferred the recipe to model families it initially failed on: *spectral-PCA* (per-head matrix- Γ replacement of post-projection norms in Qwen3), *pair-PCA* (RoPE-commuting $SO(2)$ rotation per RoPE pair in non-norm architectures), and *MoE-aware* input-side absorption (handling fused 3D experts in modern HF MoE layouts).
- (iv) A bit-width ablation at W2 vs. W4 showing the empirical payoff scales with per-channel quantization-noise budget, falling within the evaluation noise floor at W4 — consistent with the Schur-convexity intuition that the multiplicative cost of unconcentrated influence vanishes as the noise budget shrinks.
- (v) An architectural-fuzzing experiment using poolside’s Laguna-XS.2 modeling source (33B MoE, hardware-out-of-reach) that surfaced and fixed a generalization bug in our input-side attention absorption (the previously-uncovered `g_proj` input-side gate pattern).

Limitations and honest framing. This is an applied paper about an integer-rounding heuristic that empirically improves wikitext-2 PPL at extreme bit-widths. We do not claim:

- that the WHT-energy quantity defined here equals the Boolean influence of the theory paper (it does not — one is per-coordinate post-WHT activation energy, the other is Fourier mass over subsets containing the coordinate);
- that the Schur-convexity theorem of the theory paper transfers formally to real-valued layers (it doesn’t — the transfer is intuitive, not theorem);
- competitive results against the strongest published low-bit baselines (we compare against vanilla auto-round, which is itself a strong baseline; head-to-head comparison against AWQ, GPTQ, SpinQuant, and QuaRot at matched calibration is left to future work);
- statistical significance in the standard-deviation sense (results are single-seed; we treat differences below the empirical ± 0.5 PPL noise floor as no-effect).

The contribution is engineering value: a small, principled, exactly-math-invariant pre-quantization transformation with non-trivial empirical wins at W2 across multiple model families.

Position relative to rotation- and smoothing-based quantization. Several recent lines of work also use orthogonal rotations or per-channel rescaling to ease low-bit quantization. *SmoothQuant* [10] migrates per-channel activation outliers into the weights via a diagonal scale factor, the structural ancestor of the per-coordinate scaling used here. *AWQ* [2] chooses similar scales by a salience criterion derived from activation magnitude. On the rotation side, *QuaRot* [5] applies fixed Hadamard rotations to suppress activation outliers for INT4 inference; *SpinQuant* [4] learns the rotation jointly with the quantizer at W4A4 / W4A8; and most directly adjacent to this work, *ButterflyQuant* [16] learns a butterfly-structured orthogonal transform via Givens-angle gradient descent (128 calibration samples, $O(n \log n)$ trainable parameters), arguing that fixed Hadamard transforms “cannot adapt to specific weight distributions.” The strongest 2-bit weight-only baselines are *QuIP#* [12] (random Hadamard incoherence + E8 lattice codebook), *AQLM* [13] (additive multi-codebook), and *OmniQuant* [11] (learnable weight clipping + equivalent transformation). *KVQuant* [15] addresses the RoPE-key quantization difficulty at long context.

This paper differs in motivation and use: (i) the rotation is the *fixed* Walsh-Hadamard butterfly — no learning, no outlier statistics, no codebook; (ii) per-channel adaptivity is recovered not by learning the rotation but by a separate per-coordinate energy scaling $s_\ell = (1 + d\rho_\ell)^\alpha$ inspired by the Boolean-influence Schur-convexity of the companion theory paper [1], biasing per-group rounding budget toward high-spectral-energy channels — a direct response to the “fixed transforms cannot adapt” position of [16]: the adaptation lives in the diagonal scale, not the orthogonal factor; (iii) the regime targeted is W2A16, where the quantization noise budget is large and per-channel allocation dominates over outlier flattening; and (iv) the per-head spectral-PCA and SO(2)-pair-PCA extensions described below resolve the q_norm /RoPE compatibility failures that arise when a rotation enters the attention block — a regime that QuaRot, SpinQuant, and ButterflyQuant do not engage with directly. Head-to-head benchmarking against SpinQuant, QuaRot, QuIP#, AQLM, OmniQuant, and ButterflyQuant at matched calibration is the main item in our future-work list.

2 Method

2.1 Real-valued spectral-energy proxy

For a layer $L : \mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^{d_{\text{out}}}$ with input activations x collected on a calibration set, define the BBT spectral-energy proxy of input coordinate ℓ as

$$\rho_\ell(L) = \frac{\mathbb{E}[h_\ell^2]}{\sum_k \mathbb{E}[h_k^2]}, \quad h = H_{d_{\text{pad}}} x_{\text{pad}} / \sqrt{d_{\text{pad}}}, \quad (1)$$

where x_{pad} zero-pads x to the next power of two d_{pad} and $H_{d_{\text{pad}}}$ is the Walsh-Hadamard matrix. The zero padding is used only to make the WHT dimension a power of two; although the padded entries of x_{pad} are zero before rotation, the WHT mixes energy across all d_{pad} spectral coordinates, so ρ_ℓ is well-defined on the full padded index set. In implementation the weight is temporarily embedded as $W_{\text{pad}} = [W \ 0] \in \mathbb{R}^{d_{\text{out}} \times d_{\text{pad}}}$, the rotated and scaled weight is formed in the d_{pad} -dimensional spectral basis, and the inverse/folding step returns an ordinary $d_{\text{out}} \times d_{\text{in}}$ dense layer — no padded input coordinate is introduced into the deployed model. We reserve the symbol $\text{Inf}_\ell(f) = \sum_{S \ni \ell} \hat{f}(S)^2$ for the Boolean influence of the theory paper [1] and use $\rho_\ell(L)$ for the real-valued per-coordinate WHT activation-energy proxy used here. The two are distinct mathematical

objects — the Boolean influence is a Fourier-mass quantity over subsets containing ℓ , while ρ_ℓ is a per-coordinate energy after a WHT rotation — but they play the same downstream role (deriving per-channel scales $s_\ell = (1 + d_{\text{pad}} \rho_\ell)^\alpha$). We do not claim formal equivalence.

2.2 Quantization recipe

We use row-vector activation notation throughout, matching PyTorch’s $\text{F.linear}(x, W) = xW^\top$. Write $Q = H_{d_{\text{pad}}} / \sqrt{d_{\text{pad}}} \in \text{O}(d_{\text{pad}})$ (orthogonal) and $D = \text{diag}(s)$ with $s_\ell = (1 + d_{\text{pad}} \cdot \rho_\ell(L))^\alpha$, $\alpha = 0.5$, following AWQ’s per-channel pre-scaling structure [2]. The transformed layer is

$$\tilde{x} = x_{\text{pad}} Q D^{-1}, \quad \tilde{W}^\top = D Q^\top W_{\text{pad}}^\top, \quad (2)$$

so that at full precision

$$\tilde{x} \tilde{W}^\top = x_{\text{pad}} Q D^{-1} D Q^\top W_{\text{pad}}^\top = x_{\text{pad}} W_{\text{pad}}^\top = x W^\top, \quad (3)$$

i.e. an exact identity (verified to relative error $< 5 \times 10^{-7}$ on randomised inputs of width 576). Operationally:

- **Spectral basis (no extra graph operator).** The rotation Q and scale D are folded into the stored weight $\tilde{W}$; a per-layer forward pre-hook applies the matching $x_{\text{pad}} Q D^{-1}$ to the input. High- ρ_ℓ columns are boosted before quantization and divided back after.
- **Quantization.** Run Intel auto-round W2A16 SignRound [3] on $\tilde{W}$ with group size 64. The boosted high-energy columns receive proportionally more grid resolution from the per-group quantizer; un-folding Q and D at dequantization preserves the math.

2.3 Models and metric

We evaluate on wikitext-2 perplexity (teacher-forced, 8 192 tokens, sliding window 1 024) for four pretrained decoder-only models: SmoLLM-135M, SmoLLM-360M (Llama-style; HuggingFaceTB), and Qwen2.5-0.5B, Qwen2.5-1.5B (Qwen2 architecture; Alibaba). All have grouped-query attention with tied input/output embeddings. Quantization runs on Intel Arc B580 (12 GB) under WSL Ubuntu; results are validated end-to-end through OpenVINO IR on Intel NPU + Arc dGPU + CPU and shown to be device-invariant within ± 0.1 PPL.

3 Results

3.1 Headline: BBT-spectral at W2A16

Reproducibility. Unless otherwise noted, all W2A16 runs use Intel auto-round 0.12.2 with group size 64, SignRound, `enable_alg_ext=False`, and the same calibration-token budget (128 samples of 2,048 tokens drawn from the wikitext-2 train split, fixed seed) for vanilla and BBT variants. Perplexity is computed teacher-forced under `transformers 4.57.6` in fp16, with sequence length 1,024 over a fixed 8,192-token wikitext-2 evaluation subset (a hardware-turnaround-time choice; full validation/test evaluation is in our next-revision queue), using the archived `eval_ppl.py` in `boolean_fourier/bbt_quant/`.¹ A separate axis is auto-round’s `enable_alg_ext` (“auto-round-

¹Absolute W2 perplexity is sensitive to the exact evaluation stack, because extreme-low-bit checkpoints amplify small numerical and modeling-code differences. We therefore pin the stack (`transformers 4.57.6`, `torch fp16`, `auto-round 0.12.2`) in `requirements.txt`, and every result JSON emitted by `eval_ppl.py` records its own `transformers/torch` versions so any number can be traced to the stack that produced it. The relative BBT improvement remained stable across the stacks we checked.

Table 1: wikitext-2 PPL under W2A16 quantization with group size 64. BBT-spectral applies the WHT rotation and influence-derived per-channel scaling *before* auto-round’s reconstruction-error optimization runs; the spectral basis biases the quantizer’s per-group dynamic range toward high-influence channels.

Model	Params	d_h	d_{ff}/d_h	Vanilla	BBT-spectral	Δ
SmolLM-135M	134.5 M	576	2.67	81.03	45.11	−44.3%
SmolLM-360M	361.8 M	960	2.67	43.21	36.55	−15.4%
Qwen2.5-0.5B	494 M	896	5.43	119.31	50.22	− 57.9%
Qwen2.5-1.5B	1.54 B	1536	5.83	36.93	28.16	−23.7%

best”) flag: we report the headline with it off; enabling it strengthens the vanilla baseline substantially (roughly halving its W2 PPL) and narrows—but does not eliminate—the BBT margin. Reported values are single-seed; based on repeated calibration/evaluation runs on representative models we treat differences below approximately ± 0.5 PPL as evaluation noise. The exponent α in the per-channel scale $s_\ell = (1 + d_{\text{pad}} \rho_\ell)^\alpha$ is fixed at $\alpha = 0.5$ throughout (we deliberately do not per-model-tune α to keep the recipe closed-form).

Table 1 reports vanilla W2A16 (auto-round only) versus BBT-spectral W2A16 (auto-round on the WHT-rotated, influence-scaled weights) for each model.

Findings.

- (i) **BBT-spectral reduces wikitext-2 PPL on every model tested**, with reductions ranging from 15% to 58%. The Walsh-basis structural intuition — that per-coordinate spectral mass identifies channels worth preserving more carefully under coarse rounding — holds qualitatively in the real-valued LLM setting at extreme quantization (W2A16).
- (ii) **The relative win correlates with quantization noise headroom, not parameter count.** Qwen2.5-0.5B (494 M params) has both the worst vanilla baseline (PPL 119.31, attributable to its wide MLP factor $d_{ff}/d_h = 5.43$) and the largest BBT win (−57.9%). SmolLM-360M, with the best vanilla baseline relative to its scale (PPL 43.21), has the smallest BBT win (−15.4%). The mechanism is consistent: BBT redistributes per-group quantization error from high-influence to low-influence coordinates — when the total error budget is large, the redistribution payoff is large.
- (iii) **Architectural width correlates with BBT effect size.** The two model families differ in d_{ff}/d_h ratio: SmolLM has 2.67, Qwen2.5 has 5.4–5.8. At comparable parameter counts (SmolLM-360M $\approx$ Qwen2.5-0.5B), the wider-MLP Qwen has a $2.8\times$ worse vanilla W2 baseline (PPL 119 vs 43); BBT’s relative win is also $3.8\times$ larger (−57.9% vs −15.4%). One possible explanation is that wider intermediate states create more uneven per-coordinate ρ_ℓ profiles, increasing the value of per-channel reallocation. We do not directly prove this here: a concentration index of ρ_ℓ (e.g. Gini or HHI) per model versus the BBT win, plus a controlled width ablation, would be the cleanest follow-up ablation.
- (iv) **All gains validated through OpenVINO IR.** Each BBT-spectral checkpoint was dequantized to FP16, exported to OpenVINO IR, and evaluated on Intel NPU (Core Ultra integrated AI Boost), Intel Arc B580 dGPU, and CPU. PPL is invariant to deployment device within ± 0.1 on every model, confirming the gains are weight-level (not numerical artifacts of the quant runtime).

3.2 Per-head spectral PCA for post-projection normalization (Qwen3)

Qwen3-0.6B (model_type=qwen3, GQA 16:8, $d_h = 128$) introduces per-head RMSNorm layers (q_norm, k_norm) immediately after q_proj and k_proj , before RoPE and attention. Standard input-side BBT-spectral degrades it badly (vanilla 136.76 vs. BBT-spectral $> 10^7$): the per-head $\sqrt{\mathbb{E}[y^2]}$ denominator is sensitive to the per-output-channel quant-noise asymmetry that BBT’s column scaling introduces. We resolve this with a per-head reformulation we call *spectral-PCA*:

1. Calibrate the per-head output covariance $C_h = \mathbb{E}[y_h y_h^\top]$ where $y_h \in \mathbb{R}^{d_h}$ is the h -th head slice of the q_proj (or k_proj) output.
2. Eigendecompose $C_h = U_h \Lambda_h U_h^\top$ and rotate the projection’s per-head row block: $W_q[h] \leftarrow U_h^\top W_q[h]$.
3. Replace q_norm with a MATRIXGAMMARMSNORM whose forward applies the per-head matrix $\Gamma_h = \text{diag}(\gamma) U_h$ (instead of the original elementwise γ). One can verify $\Gamma_h (U_h^\top y_h) / \text{RMS}(U_h^\top y_h) = \gamma \odot y_h / \text{RMS}(y_h)$, i.e. the post-norm output equals the unmodified-model output *exactly* at full precision; RoPE and attention downstream see vanilla inputs and the math is invariant end-to-end (verified numerically: relative logits error 6×10^{-6}).
4. Under GQA, query heads sharing a kv-head must use the kv-head’s U so the post-RoPE dot product preserves; we replicate $U_{kv}[h_{kv}]$ across the q -heads in each kv-group.

At quantization time, auto-round rounds INT2 values for the *rotated* W_q rows. Because the row dynamic range in the eigenbasis is concentrated in a few principal directions, INT2 group-quant noise after un-rotation ($U_h \epsilon$) is aligned with the activation’s natural variance directions, which q_norm’s RMS denominator absorbs gracefully rather than amplifying. Empirically:

Variant	Wikitext-2 PPL
Vanilla auto-round W2 g64	136.76
BBT no-rotation (norm-absorbed)	165.49
BBT-spectral (input-side WHT)	$> 10^7$
BBT spectral-PCA (Route A)	88.99

The spectral-PCA variant beats the vanilla baseline by **35%**, demonstrating that the previously-named “architectural boundary” was an implementation gap rather than a theoretical limit.

A subtle point: the natural “symmetric” choice $\Gamma_h = U_h^\top \text{diag}(\gamma) U_h$ (which makes the post-norm output equal $U_h^\top$ times the vanilla output) breaks under RoPE because RoPE applies a position-dependent rotation R_t that does not commute with arbitrary U_h . Specifically $(R_t U_h^\top q) \cdot (R_s U_h^\top k) \neq (R_t q) \cdot (R_s k)$ unless $t = s$. The asymmetric choice $\Gamma_h = \text{diag}(\gamma) U_h$ makes the post-norm output equal vanilla *directly*, so RoPE then receives identical inputs to the unmodified model. Empirically the symmetric choice gave PPL ≈ 18886 (math broken); the asymmetric choice gave 88.99.

3.3 Pair-PCA: extending Route A to non-q_norm architectures

For SmolLM, Qwen2.5, and Llama-style architectures (no q_norm/k_norm to absorb a per-head matrix), the orthogonal rotation must instead commute with RoPE. RoPE acts as a 2D rotation in each plane $(i, i + d_h/2)$ for $i \in [0, d_h/2)$, so the only orthogonal matrices on $\mathbb{R}^{d_h}$ commuting with all RoPE rotations are products of independent SO(2) rotations on these planes — a much smaller group of dimension $d_h/2$ rather than $d_h(d_h - 1)/2$. The constrained PCA computes a 2×2

covariance per (head, RoPE-pair), eigendecomposes to $U_{h,p} \in \text{SO}(2)$ (forcing $\det = +1$, since `eigh` can return reflections in $\text{O}(2)$ which do *not* commute with rotations), and applies the rotation to the corresponding row-pair of W_q / W_k (and bias). At dequant we un-rotate per-pair, restoring a plain unmodified-architecture model. Empirical results, layered on top of input-side norm-absorbed BBT (W2A16, group size 64, $\alpha = 0.5$):

Model	Vanilla	BBT-spectral (input-WHT)	pair-PCA	vs. best prior
SmolLM-135M	81.03	45.11	61.37	+36% worse
SmolLM-360M	43.21	36.55	30.34	−17%
Qwen2.5-0.5B	119.31	50.22	40.15	−20%
Qwen2.5-1.5B	36.93	28.16	21.84	−22% (−41% vs. vanilla)

Pair-PCA improves over input-WHT BBT-spectral on three of four models (SmolLM-360M, Qwen2.5-0.5B, Qwen2.5-1.5B) by 17–22%, with the largest models showing the largest gains. It underperforms on the smallest SmolLM-135M (where the constrained $\text{SO}(2)^{d_h/2}$ search space leaves less room than the d_h -wide WHT input rotation, and per-pair covariance estimation is noisier with fewer attention activations). The recipe is a strict generalization rather than a replacement: it composes with input-side BBT and is selected per-model.

3.4 Bit-width scaling: redistribution payoff is a function of the noise budget

A natural reviewer question is whether the per-channel redistribution survives at higher bit-widths. We ran the same recipe on Qwen2.5-1.5B at W4A16 $g=64$ with both vanilla auto-round and BBT pair-PCA:

Bits	Vanilla	BBT pair-PCA	Δ
W2	36.93	21.84	−41%
W4	9.65	9.71 +0.06 (within noise)	

At W4 the per-channel quantization noise has already collapsed to near-zero on every channel; vanilla auto-round is within $\sim 1\%$ of the FP16 ceiling and there is no longer a noise budget to redistribute. The Schur-convexity intuition (theory paper [1], Theorem 3) is precisely about the *multiplicative cost of unconcentrated influence under a fixed noise budget*: when the budget shrinks, the cost shrinks with it. The 0.06 PPL spread at W4 sits inside our wikitext-2 evaluation noise floor (single-seed runs have ± 0.5 PPL variance from calibration sampling) and we report it as a no-effect data point. We emphasize this as a deliberate negative control: when the quantization noise is small, the proposed redistribution has no measurable benefit. The method does not “magically improve everything” — its operating regime is the high-noise extreme-low-bit corner where vanilla per-group rounding leaves a large noise budget to redistribute. BBT-quantization’s value, then, is concentrated at extreme low-bit regimes (W2) where the absolute quantization cost of vanilla per-group rounding is high enough that any per-channel reweighting of that cost has measurable downstream impact. We did not extend the ablation to W3 because auto-round’s 3-bit packing is irregular (the bit-width does not divide 32 evenly) and our INT2/INT4-style unpacker does not implement it; testing W3 is left to future work.

3.5 Scale-exponent ablation

We fix $\alpha = 0.5$ throughout the headline results to keep the recipe closed-form, but a reviewer will reasonably ask whether that choice is load-bearing. Table 2 sweeps $\alpha \in \{0, 0.25, 0.5, 1.0\}$ on Qwen2.5-0.5B at W2A16 $g=64$ (input-side BBT-spectral; same eval stack as the headline, `transformers`

4.57.6, 8,192 tokens, window 1,024; here with auto-round `enable_alg_ext on`, so the vanilla baseline is the stronger 70.41 rather than the headline’s 119.31).

Table 2: Scale-exponent ablation on Qwen2.5-0.5B (W2A16, BBT-spectral). $\alpha = 0$ is the fixed WHT rotation with *no* per-channel scaling. The optimum is shallow near $\alpha \approx 0.25$ – 0.5 ; over-aggressive scaling ($\alpha = 1$) hurts. Most of the gain comes from the rotation alone, with per-channel energy scaling adding a smaller further improvement.

Variant	Wikitext-2 PPL	vs. vanilla
Vanilla (auto-round, <code>alg_ext on</code>)	70.41	—
BBT $\alpha = 0$ (rotation only)	63.18	−10.3%
BBT $\alpha = 0.25$	51.64	− 26.7%
BBT $\alpha = 0.5$ (default)	51.74	−26.5%
BBT $\alpha = 1.0$	57.53	−18.3%

Two observations. First, $\alpha = 0.5$ is within noise of the empirical optimum $\alpha \approx 0.25$, so the fixed closed-form choice costs essentially nothing; aggressive scaling ($\alpha = 1$) is clearly worse. Second, the fixed rotation alone ($\alpha = 0$) already captures a meaningful fraction of the gain, with the per-channel energy scaling contributing the remainder — consistent with the basis/scale factorization in the discussion, where Q and D play complementary rather than redundant roles.

3.6 Additional architectural data point: TinyLlama-1.1B

TinyLlama (Llama-family, GQA 32:4, hidden 2048, intermediate 5632) at W2A16 $g=64$ with BBT pair-PCA gives wikitext-2 PPL 16.0. The vanilla baseline could not be obtained on our hardware (Arc B580 12 GB) because the no-rotation path runs at `batch_size=4` and the wider 5632-dim intermediate combined with the per-iter activation footprint pushed peak VRAM past the SYCL spill threshold; per-block tuning collapsed to several hours. The BBT pair-PCA path uses `batch_size=1` (the rotation-mode auto-round patch noted in the implementation notes) and stays under the threshold, completing in ~ 25 minutes. The PPL 16.0 result is in the same regime as Qwen2.5-1.5B’s BBT W2 result of 21.8 at a comparable parameter scale; the lack of a vanilla baseline at this configuration is a hardware-tooling artifact rather than a methodological one. We note that pair-PCA’s inverse step at dequant time must un-rotate *both* the weight rows and the bias, since Qwen2.5 has q_proj/k_proj biases (Llama and SmolLM do not); omitting the bias inversion gives PPL $\approx 35\,769$, a useful diagnostic for the math-invariance check.

3.7 NPU / GPU / CPU deployment throughput

Earlier we reported that the BBT-spectral PPL is invariant to deployment device within ± 0.1 across Intel NPU (AI Boost), Intel Arc B580 dGPU, and CPU — confirming the gains are weight-level. Table 3 extends this to absolute throughput across all eleven dequantized FP16 checkpoints we built (vanilla baselines for SmolLM-135M, SmolLM-360M, Qwen2.5-0.5B; BBT-spectral, pair-PCA (`spectral_pca_2d`), and Route-A spectral-PCA variants where applicable; TinyLlama-1.1B-pair-PCA as a Llama-family data point). All measurements are on the same Core Ultra 5 225F + Arc B580 box, via `openvino_genai 2026.1 LLMPipeline` (which handles the static-shape compilation NPU requires for LLM IRs); prompt length 16, decode 64 new tokens, greedy, median of 3 runs after a 2-run warmup.

Reading the table.

Table 3: Deployment throughput across all dequantized FP16 BBT checkpoints. “FTL” = first-token latency in ms (median); “tok/s” = steady-state decode tokens/sec (median); – = not measured. Hardware: Intel Core Ultra 5 225F + Arc B580 (12 GB) + AI Boost NPU. Tokenizer for Qwen3-0.6B was patched from the vanilla checkpoint (its dir lacked `special_tokens_map.json`); no other manipulation.

Model	Variant	NPU		GPU.0 (Arc)		CPU	
		FTL (ms)	tok/s	FTL (ms)	tok/s	FTL (ms)	tok/s
SmolLM-135M	vanilla	234	49.8	16	197.6	13	110.0
SmolLM-135M	BBT-spectral	233	50.1	16	221.8	14	123.5
SmolLM-360M	vanilla	479	26.9	18	181.5	27	52.6
SmolLM-360M	BBT-spectral	476	27.4	17	186.2	23	57.9
Qwen2.5-0.5B	vanilla	462	23.4	13	198.8	27	47.5
Qwen2.5-0.5B	BBT-spectral	467	23.2	14	193.5	26	46.6
Qwen2.5-0.5B	pair-PCA	470	22.9	19	150.9	29	39.7
Qwen3-0.6B	spectral-PCA	1013	15.5	23	151.8	30	39.0
TinyLlama-1.1B	pair-PCA	1733	1.4	19	201.8	27	47.4
Qwen2.5-1.5B	BBT-spectral	2224	1.4	23	131.2	33	34.2
Qwen2.5-1.5B	pair-PCA	2254	1.4	21	136.4	33	34.3

- **BBT introduces no architectural runtime operators after folding.** Since the WHT rotation, per-channel scale, and per-head/per-pair PCA matrices are all folded into dense FP16 weights before OpenVINO export, the deployed graph is a standard dense transformer — identical operator set to the vanilla baseline. Throughput is therefore broadly comparable for matched dense checkpoints, but not identical: most variants are within roughly 10–12% of vanilla (NPU within 0.6%, CPU within 11%), while Qwen2.5-0.5B pair-PCA shows a larger GPU drop of about 24% (198.8 $\rightarrow$ 150.9 tok/s), likely from backend graph-layout or kernel-cache effects after folding. We therefore frame this as “no extra runtime mechanism,” not “no guaranteed slowdown”: backend compilation, cache effects, and graph layout produce model-dependent variation even at fixed operator structure.
- **Arc B580 dominates throughput** on every model, 4–8 $\times$ the NPU’s tokens/sec on sub-1B models. This is unsurprising: the dGPU has $\sim$ 230 TOPS at int8 and a wide memory bus; AI Boost is targeted at 11 TOPS sustained.
- **NPU memory wall at $\sim$ 1 B parameters** on this hardware. TinyLlama-1.1B and Qwen2.5-1.5B both collapse to 1.4 tok/s on NPU — consistent with the AI Boost runtime spilling parts of the weight tensor through system memory once the model exceeds the $\sim$ 4 GB on-package buffer. Sub-1B models stay on-NPU and run at the expected 15–50 tok/s. We treat 1 B as the practical cutoff for NPU deployment on this generation of silicon. GPU and CPU are unaffected by this cutoff.
- **NPU first-token latency scales linearly with model size:** 233 ms (0.135 B) $\rightarrow$ 476 ms (0.36–0.5 B) $\rightarrow$ 1013 ms (0.6 B) $\rightarrow$ 1733–2254 ms (1.1–1.5 B). Most of the cost is static-shape graph compilation and prefill staging on NPU, which subsequent token decodes amortize cleanly when within the on-package memory budget but cannot amortize once the model spills.
- **NPU vs. CPU: comparable steady-state for sub-1B.** On SmolLM-135M and Qwen2.5-0.5B the CPU is $\sim$ 2 $\times$ faster than NPU; on SmolLM-360M they are within 2 $\times$ of each other.

The NPU’s value proposition is power, not raw speed — AI Boost typically draws single-digit watts under sustained inference vs. the CPU’s 20+ watts at full load. We did not instrument power for this paper; published silicon numbers are the basis for the claim.

The table grounds the “we deploy through OpenVINO IR” claim with concrete numbers across the full BBT model zoo. A broader sweep with explicit power measurement, longer prompts, and prompt/decode ratio sweeps is left to follow-up work.

4 Implementation

The empirical pipeline was built atop Intel auto-round 0.12.2; the BBT-spectral step required (a) monkey-patching auto-round’s quantization wrapper to intercept the per-layer forward, applying the spectral pad+rotate+inverse-scale transform on the input before the F.linear call; (b) caching the device-and-dtype-cast Hadamard matrix and scale vector per layer to avoid the $\sim 200\text{-iter} \times N\text{-linear}$ allocation churn that fragments consumer XPU memory; (c) for SmolLM-360M and Qwen2.5-1.5B, reducing the auto-round inner-iter batch size (4 for vanilla, 1 for spectral, conditional on the rotation mode) to keep the per-block working set under the 12 GB Arc B580 spill threshold. These changes are non-invasive (no auto-round source modifications) and idempotent. Reference implementation and reproducibility scripts at `boolean_fourier/bbt_quant/`.

Full-precision invariance checks. Each BBT mode is paired with a per-architecture unit test that round-trips a synthetic fixture — random input through original model and through transformed-then-inverse model — and compares logits. Table 4 consolidates the numerical floors observed on Arc B580 at fp32.

Table 4: Full-precision math-invariance checks before quantization. Each row reports the relative logits (or per-layer activation) error between the original and BBT-transformed-then-inverted forward pass on a randomized synthetic fixture matching the architecture’s modeling source. Floors are dominated by fp32 round-off in the $\text{WHT} \div \sqrt{d_{\text{pad}}}$ normalization.

Transformation	Tested architecture	Relative error
Input-WHT BBT-spectral	SmolLM/Qwen2-style linear blocks	$< 5 \times 10^{-7}$
Spectral-PCA Route A (matrix- Γ)	Qwen3-0.6B q_norm + RoPE block	6×10^{-6}
Pair-PCA with bias inversion	Qwen2.5-style attention + RoPE	$\sim 1 \times 10^{-7}$
Laguna g_proj absorption (fixed)	Synthetic Laguna with q_norm + g_proj	3×10^{-7}

Architectural fuzzing via Laguna-XS.2 (poolside). We attempted to apply our toolkit to poolside’s Laguna-XS.2² — a 33B MoE model with 256 routed experts plus 1 shared expert, alternating sliding/full attention, GQA 48:8, and post-projection q_norm/k_norm (Qwen3-style). The model itself is far out of reach for our 12 GB GPU (smallest INT4 variant is ~ 14 GB and pre-quantized), but instantiating a tiny synthetic Laguna directly from the released modeling source surfaced a previously-undetected bug in our *no_rotation* Group 1 absorption. Laguna’s LagunaAttention class adds a per-head “head gate” module: `g_proj: nn.Linear(d, h)` that consumes input_layernorm output, runs through softplus, and multiplies the attention output before *o_proj*. Our absorption was scaling only $q/k/v_proj$ input columns; with *g_proj* left unscaled,

²<https://huggingface.co/poolside/Laguna-XS.2>

Table 5: Bit-equivalence of the buggy and fixed input-side absorption code paths on every architecture for which the paper reports W2A16 PPL. Tiny synthetics ($d = 64$, 2 layers, GQA 4:2) instantiated from the same HuggingFace modeling classes as the production checkpoints. “Absorbs” = Linear in `self_attn` that the corresponding code path applies the per-channel scale to. “Rel-err” = $\max |y_{\text{prep}} - y_{\text{ref}}| / \max |y_{\text{ref}}|$ in fp32. “max $|\Delta|$ ” = element-wise max difference between the buggy and fixed prepared-model logits.

Reported model	Buggy absorbs	Fixed absorbs	Rel-err (both)	max $ \Delta $
SmolLM-135M (Llama)	$\{q, k, v\}_{proj}$	$\{q, k, v\}_{proj}$	1.7×10^{-7}	0
SmolLM-360M (Llama)	$\{q, k, v\}_{proj}$	$\{q, k, v\}_{proj}$	1.7×10^{-7}	0
Qwen2.5-0.5B (Qwen2)	$\{q, k, v\}_{proj}$	$\{q, k, v\}_{proj}$	2.4×10^{-7}	0
Qwen2.5-1.5B (Qwen2)	$\{q, k, v\}_{proj}$	$\{q, k, v\}_{proj}$	2.4×10^{-7}	0
TinyLlama-1.1B (Llama)	$\{q, k, v\}_{proj}$	$\{q, k, v\}_{proj}$	1.7×10^{-7}	0
Qwen3-0.6B (Qwen3)	$\{q, k, v\}_{proj}$	$\{q, k, v\}_{proj}$	2.1×10^{-7}	0
Synthetic Laguna (positive control)	$\{q, k, v\}_{proj}$	$\{q, k, v, g\}_{proj}$	1.4×10^{-2} buggy / 3×10^{-7} fixed	$\approx 10^{-2}$

the forward pass through the prepared model evaluated $\text{softplus}((W_g \cdot x)/s) \neq \text{softplus}(W_g \cdot x)/s$, leaking the per-channel scaling through a non-linearity and breaking math invariance at fp32 precision (relative error $\approx 1.4\%$ in our test). The fix is one line — extend the input-side attention consumer list from $(q, k, v)_{proj}$ to $(q, k, v, g)_{proj}$ with an `in_features = d` filter to keep it from accidentally picking up output-side projections — and it generalizes beyond Laguna: similar input-side attention gates appear in Granite-MoE, MiniMax, and other recent variants. Regression tests on SmolLM-135M, Qwen2.5-0.5B, and a synthetic Qwen2-MoE remain at the fp32 numerical floor; the new test on synthetic Laguna with both *spectral_pca* (matrix- Γ) and *spectral_pca_2d* (pair-PCA) round-trips Laguna’s full-precision logits to relative error $\approx 3 \times 10^{-7}$. We report this not as a Laguna PPL number — the hardware barrier prevents that — but as an example of the kind of out-of-distribution architectural test that catches generalization bugs cheaply. The toolkit code path for actually quantizing Laguna is ready when sufficient compute is available.

The `g_proj` fix does not retroactively affect reported PPL. Because the fix was strictly additive — it appended `g_proj` to the input-side allowlist while keeping the existing $(q, k, v)_{proj}$ entries unchanged — it is bit-for-bit a no-op on any architecture whose `self_attn` has no Linear named `g_proj` (or analog) with `in_features = d`. We verify this directly in `boolean_fourier/bbt_quant/test_reported`, which for each reported architecture instantiates a tiny synthetic from the same HuggingFace modeling class (`LlamaForCausalLM` for SmolLM and TinyLlama, `Qwen2ForCausalLM` for Qwen2.5, `Qwen3ForCausalLM` for Qwen3-0.6B), runs `_bbt_prep_no_rotation` twice on independent copies — once with the buggy three-element allowlist $(q, k, v)_{proj}$, once with the fixed four-element allowlist $(q, k, v, g)_{proj}$ — and compares the post-prep fp32 logits. Table 5 summarizes: on every reported architecture both allowlists scale exactly the same set $\{q_{proj}, k_{proj}, v_{proj}\}$, both round-trip to vanilla logits at the fp32 floor ($\approx 2 \times 10^{-7}$ relative), and the two prepared models produce the same logits to machine zero ($\max |y_{\text{fixed}} - y_{\text{buggy}}| = 0$). The PPL numbers in Table 1, the pair-PCA Section, and Table 3 are therefore unchanged by the fix; the test acts as a regression guard against re-introducing the divergence on any future model in this set.

5 Discussion and Limitations

Where the per-channel adaptivity lives: a separable factorization. The relationship between the present work and learned-rotation methods, in particular ButterflyQuant [16], is sharper if framed as a design-space distinction. Many math-preserving pre-quantization transformations used in rotation- and smoothing-based quantization can be written, in the row-vector convention above, as

$$x \mapsto x Q D^{-1}, \quad W^\top \mapsto D Q^\top W^\top,$$

with $Q \in O(d_{\text{in}})$ orthogonal and $D = \text{diag}(s)$ diagonal. This restricted but practically important design grid separates the choice of basis from the choice of per-channel scale. Different families occupy different cells:

- **QuaRot / QuIP# / ButterflyQuant:** optimize Q (fixed Hadamard, randomized incoherence-promoting Hadamard, or learned Givens-angle butterfly respectively); leave D trivial. The per-channel adaptation is forced into the orthogonal factor.
- **SmoothQuant / AWQ:** trivial Q (identity); optimize D from activation magnitude or salience. The per-channel adaptation lives entirely in the diagonal.
- **SpinQuant:** jointly optimize Q and D via gradient descent.
- **This paper (BBT-spectral):** *fixed* $Q = H_d$ (the Walsh-Hadamard butterfly — no learning and no calibration of the rotation itself) paired with an *adaptive* D derived in closed form from a per-coordinate Walsh-basis activation-energy statistic, $s_\ell = (1 + d_{\text{pad}} \rho_\ell)^\alpha$ (the only calibration step is a single forward pass to estimate $\mathbb{E}[h_\ell^2]$), motivated by the Schur-convex influence-Banach geometry of the companion theory paper [1].

The position taken here is that for extreme low-bit weight-only quantization, the per-channel adaptation that ButterflyQuant achieves by learning Q is largely recoverable in closed form by adapting D instead and keeping Q at its fastest, most universally-supported value (the WHT). This shifts the calibration cost from gradient training of $O(d \log d)$ Givens angles to a single forward pass over a small calibration set to estimate $\mathbb{E}[h_\ell^2]$, and keeps the runtime transformation a parameter-free $O(d \log d)$ butterfly — the same kernel target QuaRot already deploys. The per-head spectral-PCA and $\text{SO}(2)$ -pair-PCA extensions (Sections above) are recoveries of D -only failure modes inside attention: the rotation interacts non-trivially with q_norm and RoPE, and the cure is to give Q a small piece of structure too — per-head U_h in the q_norm case, per-RoPE-pair $\text{SO}(2)$ in the non-norm case — chosen to be exactly the structure that the architecture’s invariances permit. We do not claim this factorization is ours — it is implicit in any Hadamard-rotation paper — only that naming the cells makes the contrast clean and identifies the obvious next experiment: a head-to-head BBT-spectral vs. ButterflyQuant comparison at W2A16 on a shared model with a shared calibration budget, isolating the Q -only vs. D -only adaptation question. We attempted this comparison but found that the ButterflyQuant code repository cited in [16] was empty (a placeholder README only) at the time of our submission; we defer the head-to-head to a future version once upstream code is released or a faithful third-party reimplementation appears, rather than risk the comparison being confounded by reimplementation choices on our end.

What this paper does and does not claim. This is an applied paper. The contribution is engineering value: a small, principled, exactly-math-invariant pre-quantization transformation with non-trivial empirical wins at W2 across multiple model families. We do *not* claim:

- formal equivalence between the WHT activation-energy quantity defined here and the Boolean influence of the theory paper [1] — they share the same Walsh basis and the same downstream role (deriving per-channel scales) but are different mathematical objects;
- that the Schur-convexity theorem of the theory paper transfers as a theorem to real-valued layers — the transfer is heuristic, not formal;
- head-to-head superiority over AWQ, GPTQ, SpinQuant, or QuaRot at matched calibration — our comparator is vanilla auto-round, which is a strong baseline but not all of them;
- statistical significance — results are single-seed and we treat differences below the empirical ± 0.5 PPL noise floor as no-effect;
- evaluation completeness — perplexity is computed on an 8,192-token wikitext-2 subset (a hardware-turnaround-time choice), not the full validation/test split, and we do not report FP16 reference perplexities or a gap-to-FP16-closure metric in this revision;
- a swept α ablation — we fix $\alpha = 0.5$ throughout, intentionally, to keep the recipe closed-form; per-model tuning of α may improve the headline numbers and is not used here.

Future work. The next-revision queue includes: full wikitext-2 test-set evaluation plus an FP16 reference column and gap-to-FP16-closure metric on every model; multi-seed ablations against the standard low-bit quantization baseline suite (AWQ/GPTQ/SpinQuant/QuaRot/ButterflyQuant/QuIP#/AQLM/C) at matched calibration token budget; an α and group-size sweep; downstream-task evaluation (MMLU, ARC, HellaSwag, GSM8K) beyond perplexity; concentration-index (ρ_ℓ Gini/HHI) measurements per model versus BBT win; and full quantization runs of DeepSeek-V4 and Laguna-XS.2 on cloud H100/H200 instances where the per-block working set fits.

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